Speech enhancement implementation using Weiner Filtering and Spectral Subtracting

BECE207L Random Processes

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1. ABSTRACT

Speech is something that people use to talk to each other and to technology. In the world, speech is the primary mode of communication between humans and machines. However, in real-world environments, speech signals are often degraded by background noise, electronic interference, and other disturbances. This degradation reduces the clarity and intelligibility of speech, making communication difficult for both humans and automated systems.

Speech enhancement techniques are used to improve the quality of speech signals by reducing unwanted noise while preserving important speech components. In this project, two classical techniques—spectral subtraction and Wiener filtering—are implemented and analysed.

Spectral subtraction operates in the frequency domain and removes estimated noise from the noisy signal. Although effective, it may introduce artefacts. Wiener filtering, on the other hand, is a statistical approach that minimises the mean square error and produces smoother output.

A hybrid approach combining both techniques is proposed in this work. Spectral subtraction is first applied to remove major noise components, followed by Wiener filtering to refine the signal further. Fast Fourier Transform (FFT) and inverse FFT are used for signal processing.

The results demonstrate that the hybrid method improves speech clarity and reduces noise effectively. This project highlights that combining classical techniques can provide efficient and practical solutions for speech enhancement.

2. INTRODUCTION

We all talk to each other every day, and speech is the way for us to communicate. In today's technology, we use speech in things like mobile phones, voice assistants, online meetings and machines that can recognise what we say. Speech can get messed up by noise when we record or send it, which makes it hard to understand.

Noise in speech can come from places like the environment, electronic devices or other people talking. This noise mixes with the speech and makes it difficult to hear. To fix this, we use speech enhancement techniques to reduce the noise and make the speech clearer.

Making speech clearer is a job because speech is always changing. It does not stay the same, so we need to use methods that can adapt to these changes. Some old methods, like subtraction and Wiener filtering, have been used a lot because they are easy to use and work well.

This project is about trying out these methods and analysing them. We want to combine the parts of both methods to make them work better. Our goal is to make speech sound better and not get distorted.

3. OBJECTIVES

The main goal of this project is to create a speech enhancement system. This system should make speech clearer by reducing noise and keeping the parts of the sound.
Some specific goals are:

- Use subtraction and Wiener filtering methods
- See how well they work
- Compare how effective they are
- The project also wants to try combining both methods to get better results.
- Another goal is to test the system in a way. This includes:
- Checking the system's performance with numbers like Signal-to-Noise Ratio (SNR)
- Getting a feel for how well the system works
- The project also wants to show how speech enhancement can be used in real-life situations.
- The speech enhancement system is being developed to improve speech clarity.
- The system will use subtraction.
- The system will use Wiener filtering.
- The Signal-, to-Noise Ratio (SNR) will be used.

4. RESEARCH GAP

There are ways to make speech sound better, but some problems remain. A lot of methods have trouble dealing with noise that changes quickly. One simple way to fix this is called subtraction, but it can make the speech sound funny and ruin the quality. Another way is called Wiener filtering, which makes the sound smoother. It needs to know exactly what the speech and noise sound like.

One big issue is that many systems use one method at a time, rather than using them together. This makes them not work well as they could. We need to find ways to combine these methods so they can work together.

This project is trying to fix these problems by combining subtraction and Wiener filtering. The goal of this project is to make speech sound better by reducing noise and not making it sound distorted. Speech enhancement techniques like subtraction and Wiener filtering are used in this project to achieve this goal. The project aims to find a balance between being simple and working well using speech enhancement techniques, like spectral subtraction and Wiener filtering.

5. PROBLEM STATEMENT

When we record speech in the world, it gets affected by all sorts of noise. This includes things like background sounds, people talking, and interference from electronics. All this noise makes it really hard to hear what the person is saying. It is especially bad in things like phone networks and apps that use voice. When the sound is not clear, it is frustrating for the person listening. It makes the system not work very well. Speech recognition systems need the sound to be clear. When there is noise, they do not work as well.

The problem is even harder because speech sounds are always changing. The noise can also change, which makes it hard to tell what is speech and what is noise. The old ways of filtering out noise do not work well because they can get rid of important parts of the speech sound along with the noise.

So the main goal is to make a system that can reduce noise without hurting the speech sound. The speech signal system needs to be fast, accurate and able to handle various kinds of noise. It also needs to work like in real-time applications. The big challenge is finding a balance between reducing noise and keeping the speech sound. This is what we are trying to solve with the speech signal system.

6. CONTRIBUTIONS

This project presents significant contributions in the field of speech enhancement using classical signal processing techniques. One of the primary contributions is the implementation of spectral subtraction, which is a simple yet effective method for reducing noise in speech signals. This method helps in understanding how noise can be estimated and removed in the frequency domain.

Another major contribution is the implementation of Wiener filtering. This technique improves the quality of the speech signal by minimising the mean square error between the clean and noisy signals. It provides smoother and more natural output compared to basic filtering methods.

A key contribution of this project is the development of a hybrid approach that combines both spectral subtraction and Wiener filtering. This combination allows the system to remove a large portion of noise initially and then refine the signal further, resulting in improved speech clarity.

The project also includes both qualitative and quantitative analysis. Signal-to-noise ratio (SNR) is used as a quantitative metric to evaluate performance. Overall, the project demonstrates an efficient and practical approach for speech enhancement.

7. LITERATURE SURVEY

Speech enhancement has been extensively studied over the years, leading to the development of various techniques. Spectral subtraction was one of the earliest methods introduced for noise reduction. It operates in the frequency domain and is widely used due to its simplicity and effectiveness.

Wiener filtering is another classical technique that is based on statistical estimation. It minimises the mean square error between the original and processed signals. It is widely used in signal processing applications due to its effectiveness in reducing noise.

Recent advancements include machine learning and deep learning methods. These methods use neural networks to learn patterns in speech and noise. While they provide better results, they require large datasets and computational power.

Classical methods remain relevant due to their efficiency and ease of implementation. This project builds upon these techniques and combines them to achieve better results.

8. FUNDAMENTALS OF SPEECH SIGNALS

- Speech signals are time-varying signals produced by the human vocal system. They are considered non-stationary because their properties change over time. This makes processing them more complex compared to stationary signals.

- Speech consists of voiced and unvoiced components. Voiced sounds are periodic and are produced by the vibration of the vocal cords. Unvoiced sounds are noise-like and are produced without vocal cord vibration.

- To analyse speech effectively, it is divided into short frames. This allows the signal to be treated as stationary within each frame. Both time-domain and frequency-domain analyses are used.

- Understanding these fundamentals is essential for designing speech enhancement systems. It helps in distinguishing speech from noise and applying appropriate processing techniques.

9. NOISE IN SPEECH SIGNALS

- Noise is any unwanted signal that interferes with the original speech signal. It reduces the clarity and intelligibility of speech. Noise can originate from environmental sources, electronic devices, or other speakers.

- Different types of noise include white noise, Gaussian noise, and environmental noise. Each type has different characteristics and affects the speech signal differently.

- Noise affects both amplitude and frequency components of speech. It can mask important features, making it difficult to understand the signal.

- Effective noise reduction techniques are required to improve speech quality. These techniques must remove noise without affecting the original speech signal.

10. SPEECH ENHANCEMENT TECHNIQUES OVERVIEW

- Speech enhancement techniques aim to improve the quality of speech signals by reducing noise. These techniques include spectral subtraction, Wiener filtering, adaptive filtering, and machine learning methods.

- Spectral subtraction is simple and effective, but may introduce artefacts. Wiener filtering provides smoother output but requires accurate noise estimation.

- Modern techniques such as deep learning provide better performance but require more computational resources.

- Each method has its advantages and limitations. The choice depends on the application and available resources.

11. SPECTRAL SUBTRACTION METHOD

Spectral subtraction is a widely used technique for noise reduction. It works in the frequency domain and assumes that noise can be estimated and removed.

$$X(f) = Y(f) - N(f)$$

- The process involves converting the signal to the frequency domain, estimating noise, subtracting it, and converting back to the time domain.

- This method is simple and effective for stationary noise. However, it may introduce artefacts such as musical noise.

Despite its limitations, spectral subtraction is widely used due to its simplicity.

12. WIENER FILTERING METHOD

Wiener filtering is a statistical method used for speech enhancement. It minimises the mean-square error between the clean and noisy signals.

$$H(f) = \frac{S(f)^2}{S(f)^2 + N(f)^2}$$

It adjusts frequency components based on the signal-to-noise ratio. Noisy components are reduced while speech components are preserved. This method provides smoother output compared to spectral subtraction. However, it requires an accurate estimation of noise. Wiener filtering is widely used in many signal processing applications.

13. PROPOSED HYBRID METHODOLOGY

The proposed methodology combines spectral subtraction and Wiener filtering. This approach improves overall performance. Spectral subtraction removes the major portion of noise. Wiener filtering refines the signal and reduces residual noise. This combination reduces artefacts and improves speech quality. It provides a balance between noise reduction and signal preservation. The hybrid approach is more effective than individual methods and is suitable for practical applications.

14. SYSTEM ARCHITECTURE

- The system architecture includes input acquisition, preprocessing, FFT, noise estimation, filtering, and output stages.

- The noisy signal is divided into frames and converted to the frequency domain. Noise estimation is performed, followed by spectral subtraction.

- Wiener filtering is applied to refine the signal. The processed signal is converted back to the time domain. The final output is an enhanced speech signal with reduced noise.

15. FLOWCHART

15.1 Description

The flowchart shown below represents the step-by-step process used in the proposed speech enhancement system. It clearly explains how the noisy speech signal is processed through different stages, such as framing, frequency transformation, noise estimation, and filtering.

Initially, the input speech signal is taken and converted into a suitable digital format. Since speech is a non-stationary signal, it is divided into small frames to make processing easier. Each frame is then windowed to reduce signal discontinuities. After that, Fast Fourier Transform (FFT) is applied to convert the signal into the frequency domain.

In the next stage, the noise present in the signal is estimated. Spectral subtraction is then applied to remove the major portion of noise. However, since some noise may remain, Wiener filtering is used to further refine the signal and improve its quality.

Finally, the processed signal is converted back into the time domain using inverse FFT. The frames are combined to reconstruct the final enhanced speech signal. This step-by-step process ensures effective noise reduction while maintaining the quality of speech.

15.2 Flowchart Diagram

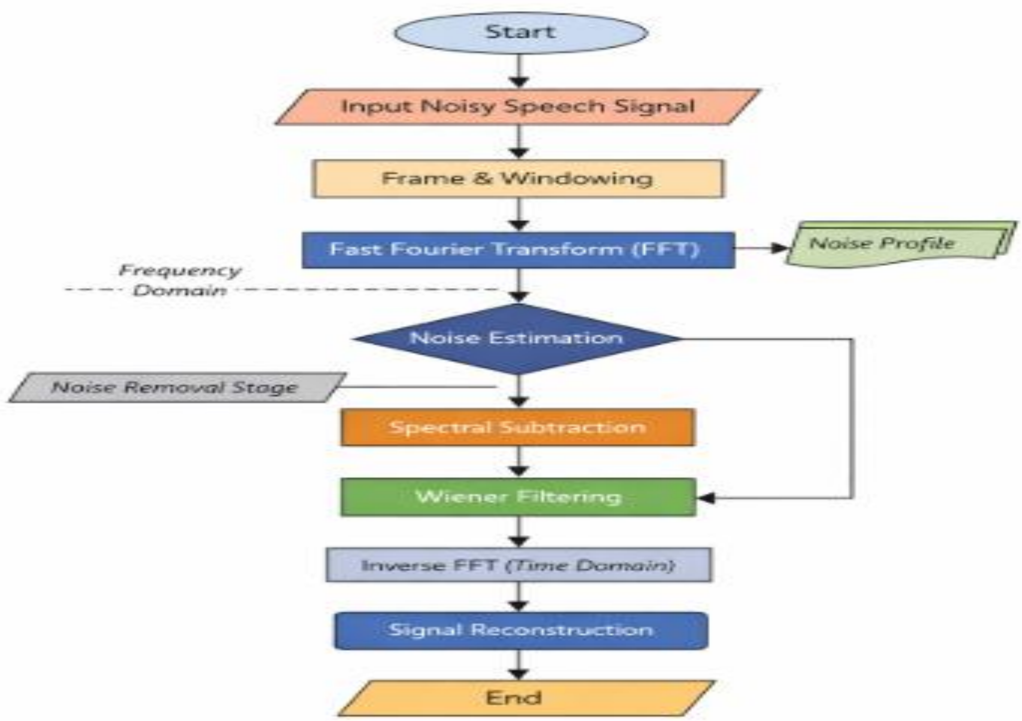

Figure 15.1: Flowchart of Speech Enhancement using Spectral Subtraction and Wiener Filtering

15.3 FLOW CHART STRUCTURE

Start
↓
Input Signal
↓
Framing + Windowing
↓
FFT
↓
Noise Estimation
↓
Spectral Subtraction
↓
Wiener Filtering
↓
IFFT
↓
Reconstruction
↓
Output Signal
↓
End

16. ALGORITHM

1. Input noisy signal
2. Divide into frames
3. Apply FFT
4. Estimate noise
5. Apply spectral subtraction
6. Apply Wiener filtering
7. Apply inverse FFT
8. Combine frames
9. Output enhanced signal

17. MATHEMATICAL MODELING

Mathematical modelling is important in understanding how speech enhancement works. It gives a clear idea of how noise affects the signal and how it can be reduced using equations. In this project, two main models are used: spectral subtraction and Wiener filtering. These models help in processing the speech signal in the frequency domain.

In spectral subtraction, the idea is simple. The noise present in the signal is estimated and then removed from the noisy speech. This can be written as:

$$X(f) = Y(f) - N(f)$$

Here, the noisy signal is reduced by subtracting the noise component. If the noise is estimated correctly, the result will be closer to the original clean speech.

Wiener filtering works in a slightly different way. It uses a statistical approach to reduce noise and improve the signal quality. The equation is:

$$H(f) = \frac{S(f)^2}{S(f)^2 + N(f)^2}$$

This formula helps in deciding how much of the signal should be kept and how much noise should be removed. Together, these models form the base of the speech enhancement system.

18. IMPLEMENTATION DETAILS

18.1 System Workflow

The proposed system is designed as a step-by-step web-based application that guides the user through the process of speech enhancement. The workflow begins with a welcome page where users can view project details such as documentation, formulas, team information, and code samples. Navigation is controlled through a sidebar, ensuring that users follow the sequence of steps without skipping important stages.

After accessing the welcome page, users proceed to the audio input stage. In this stage, users can either record their speech directly using a microphone or upload an existing audio file. The recording feature includes a time limit to ensure a manageable input size, while the upload feature supports common audio formats such as WAV and MP3.

Once the input is provided, the system processes the audio signal in the backend. If the uploaded file is in MP3 format, it is first converted into WAV format using appropriate audio processing libraries. The signal is then stored and prepared for further processing.

The system then performs speech enhancement using spectral subtraction and Wiener filtering techniques. The processed signal is analyzed, and additional operations such as noise classification and transcription are performed. Finally, the results are displayed on a dashboard where users can view waveforms, metrics, and download outputs.

18.2 Web Application Design

The web application is developed using a combination of frontend and backend technologies to ensure smooth interaction and efficient processing. The frontend is built using HTML, CSS, and JavaScript, providing an interactive user interface. The backend is implemented using Flask, which handles routing, processing, and data management.

The application consists of multiple pages, each representing a specific step in the workflow. The welcome page provides project information, while the input page allows users to record or upload audio. A loading page is used to display processing status, improving user experience.

The backend handles all core functionalities, including audio conversion, signal processing, and data storage. It also manages session states, ensuring that user inputs are preserved when navigating between pages.

The results are displayed on a dashboard using graphical representations such as waveforms and performance metrics. Libraries like Chart.js are used for visualization. The system also supports multilingual output, allowing users to view results in different languages.

Navigation is controlled using scripts to ensure that users follow the correct sequence. This structured design improves usability and ensures accurate processing of speech signals.

19. EXPERIMENTAL SETUP

1. To test the system, different speech samples are used. Clean speech signals are taken first, and then noise is added to them. This helps in checking how well the system performs in noisy conditions.

2. Different types of noise, like white noise and background noise, are used. The system processes these noisy signals and tries to improve their quality. All experiments are done using a computer with the required software and libraries installed.

3. The performance is measured using Signal-to-Noise Ratio (SNR). By comparing the input and output SNR values, we can understand how much improvement has been made. This setup helps in testing the system under different conditions and ensures that the results are reliable.

20. RESULTS AND DISCUSSION

The results show that the system is able to improve the quality of speech significantly. When noise is added to the signal, the speech becomes unclear. After applying the enhancement techniques, the noise is reduced, and the speech becomes clearer.

Spectral subtraction removes most of the noise, but sometimes small distortions remain. Wiener filtering helps in smoothing the signal and reducing these distortions. When both methods are combined, the output is better than using each method alone.

The results can be seen through graphs and by listening to the audio. The improved clarity of speech proves that the system is working effectively.

21. QUALITATIVE ANALYSIS

Qualitative analysis is based on how the speech sounds to a listener. In this project, the enhanced speech is compared with the original noisy signal by listening to both.
It is observed that the enhanced speech is much clearer and easier to understand. Background noise is reduced, and the speech sounds more natural. This is especially noticeable when using the hybrid method. However, in some cases, small distortions may still be present. These are not very noticeable and do not affect understanding much. Overall, the listening experience is greatly improved, which shows that the system is effective.

22. QUANTITATIVE ANALYSIS

- Quantitative analysis uses numerical values to measure performance. In this project, Signal-to-Noise Ratio (SNR) is used as the main metric.

- The SNR of the noisy signal is calculated first. After processing, the SNR of the enhanced signal is calculated again. The increase in SNR shows how much noise has been reduced.

- The results are usually shown in tables and graphs for better understanding. The hybrid method shows higher SNR improvement compared to individual methods.

- This analysis proves that the system is not only working visually but also performing well mathematically.

23. COMPARISON OF METHODS

In this project, spectral subtraction and Wiener filtering are compared to understand their strengths and weaknesses. Spectral subtraction is simple and easy to implement, but it may create artefacts like musical noise.

Wiener filtering gives smoother results and better quality, but it depends on accurate noise estimation. If the estimation is not correct, the performance may be reduced. The hybrid method combines both techniques. It first removes most of the noise using spectral subtraction and then refines the signal using Wiener filtering. This combination gives better results than using either method alone, making it a more reliable approach.

24. APPLICATIONS

Speech enhancement is used in many real-life applications. In mobile communication, it helps in improving call quality by reducing background noise.

- Hearing aids it allows users to hear speech more clearly even in noisy environments. Voice assistants also use speech enhancement to improve recognition accuracy.

- Other applications include online meetings, broadcasting, and security systems. In all these areas, clear speech is very important.

This project can be applied in these fields to improve communication and user experience.

25. ADVANTAGES

- The system has several advantages. It improves speech clarity, making communication easier and more effective. The methods used are simple and easy to implement.

- The system does not require high computational power, so it can be used in real-time applications. The hybrid approach provides better results by combining two techniques.

- It is also flexible and can be used in different applications. Overall, it is a practical and efficient solution for speech enhancement.

26. LIMITATIONS

Like any engineering system, the proposed speech enhancement model also has certain limitations that need to be considered. One of the major challenges is handling non-stationary noise, where the noise characteristics change rapidly over time. In such situations, accurate noise estimation becomes difficult, which directly affects the performance of both spectral subtraction and Wiener filtering techniques.

Spectral subtraction, although simple and effective, may introduce unwanted artefacts such as musical noise. These artifacts can make the output sound unnatural, especially when the noise estimation is not accurate. Similarly, Wiener filtering depends heavily on precise estimation of both signal and noise power spectra. Any error in estimation can lead to either insufficient noise removal or distortion of the speech signal.

Another limitation is the requirement of parameter tuning. Parameters such as noise estimation level, smoothing factors, and threshold values need to be carefully adjusted for different environments. This makes the system less flexible when used in varying real-world conditions.
Additionally, the system may not perform equally well for all types of noise, such as impulsive noise or highly dynamic background sounds. Computational delay can also be a concern if the system is implemented for real-time processing without optimization.

These limitations indicate that while the current system performs well under controlled conditions, there is still significant scope for improvement, especially in handling complex and dynamic noise environments.

27. SDG MAPPING

This project aligns well with the Sustainable Development Goals (SDGs) by contributing to improvements in communication technology and accessibility. One of the key SDGs supported by this work is **SDG 3: Good Health and Well-being**. By enhancing speech quality, this system can be used in hearing aids and assistive communication devices, helping individuals with hearing impairments communicate more effectively and improving their overall quality of life.

The project also supports **SDG 9: Industry, Innovation, and Infrastructure**. The development of efficient speech enhancement systems contributes to advancements in digital communication technologies. It encourages innovation in signal processing and supports the creation of smarter and more reliable communication systems.

Furthermore, the project indirectly contributes to inclusivity and equal access to technology. Clear and enhanced speech signals are essential in education, telemedicine, and emergency communication systems. By improving speech intelligibility, this work helps bridge communication gaps in real-world scenarios.

Overall, the project demonstrates how engineering solutions can contribute to global development goals by improving accessibility, promoting innovation, and enhancing quality of life through better communication systems.

28. CONCLUSION

In this project, speech enhancement techniques have been successfully implemented using spectral subtraction and Wiener filtering methods. The study focused on reducing noise present in speech signals while preserving the important characteristics of the original speech. Both techniques were analyzed individually and then combined to form a hybrid approach.

The hybrid method proved to be more effective, as it combines the strengths of both techniques. Spectral subtraction removes the major portion of noise, while Wiener filtering refines the signal and improves its overall quality. This combination results in clearer and more natural-sounding speech compared to using a single method.

The performance of the system was evaluated using both qualitative and quantitative analysis. Listening tests showed improved clarity, while Signal-to-Noise Ratio measurements confirmed the effectiveness of noise reduction.

This project demonstrates that classical signal processing methods, when used properly, can still provide strong and reliable results. It also highlights the importance of combining techniques to achieve better performance. Overall, the system provides a practical and efficient solution for speech enhancement and can be applied in various real-world applications such as communication systems and assistive devices.

29. FUTURE ENHANCEMENTS

There are many ways to improve this system in the future. One option is to use machine learning or deep learning methods for better performance. Real-time implementation can also be developed so that the system can be used in live applications. Improved noise estimation techniques can help in handling complex environments.

The system can also be integrated into mobile devices and hearing aids for practical use.

- Deep learning models (CNN, RNN)
- Real-time DSP hardware
- Mobile app integration
- Adaptive noise estimation

30. REFERENCES

The references used in this project are taken from well-established textbooks, research papers, and reliable online resources related to speech enhancement and digital signal processing. These sources provide a strong theoretical foundation as well as practical insights into techniques such as spectral subtraction and Wiener filtering. Most of the references are from IEEE papers, academic publications, and standard textbooks, which are widely accepted in the field of signal processing.

Proper citation of these sources ensures the credibility of the work and helps in understanding the background and development of speech enhancement methods. The following references were used during the preparation of this project:

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